

FUKAURA, Japan

WMO: 475740

Lat: 40.6464N

Lon: 139.9331E

Elev: 68

StdP: 100.51

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.4	-4.5	-11.7	1.4	-2.9	-10.6	1.5	-2.5	13.0	-0.1	11.7	-0.6	6.3	260	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	28.9	24.1	27.6	23.5	26.6	23.0	24.9	27.5	24.3	26.8	23.6	26.0	3.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.2	26.5	23.5	18.5	26.0	22.8	17.7	25.4	76.5	27.7	74.0	26.8	71.3	26.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.6	-6.8	31.5	1.1	1.4	-7.6	32.5	-8.2	33.3	-8.8	34.1	-9.6	35.1		
(5)			-7.8	26.1	1.2	1.0	-8.6	26.8	-9.3	27.4	-9.9	28.0	-10.8	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-0.2	0.3	3.2	8.5	13.7	17.6	21.9	23.5	19.7	13.8	7.8	2.4		
	DBStd	8.65	2.43	2.87	3.10	3.35	3.02	2.43	2.33	2.16	2.72	2.80	3.56	3.05		
	HDD10.0	1195	316	273	212	68	4	0	0	0	0	4	83	235		
	HDD18.3	2999	574	506	469	295	146	41	1	0	15	143	316	493		
	CDD10.0	1590	0	0	1	23	119	229	369	419	291	121	18	1		
	CDD18.3	353	0	0	0	0	3	20	112	161	56	1	0	0		
	CDH23.3	1583	0	0	0	0	6	28	441	911	197	1	0	0		
	CDH26.7	230	0	0	0	0	0	1	53	148	28	0	0	0		
(14) Wind	WSAvg	3.3	4.5	4.3	4.0	3.4	2.9	2.2	2.3	2.2	2.5	3.0	3.7	4.4		
(15) Precipitation	PrecAvg	1512	101	82	81	91	119	95	150	165	169	162	159	136		
	PrecMax	1898	204	147	157	150	218	217	337	322	422	355	392	232		
	PrecMin	1098	47	27	30	28	62	10	33	57	52	56	84	53		
	PrecStd	214	36	32	29	33	42	50	70	70	81	63	57	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	11.2	14.4	20.3	24.1	25.8	30.0	31.0	29.3	22.3	18.3	13.1		
		MCWB	5.8	7.8	7.9	12.6	16.0	19.8	24.6	24.9	23.6	18.2	13.9	10.5		
	2%	DB	5.9	8.1	11.9	17.8	21.9	24.1	28.1	29.6	27.2	20.8	16.4	10.6		
		MCWB	3.8	5.4	7.9	11.8	15.2	18.9	23.5	24.6	22.9	16.8	13.1	8.4		
	5%	DB	4.5	6.0	10.0	15.6	20.1	22.8	26.8	28.2	25.4	19.6	15.0	8.7		
		MCWB	2.3	3.6	7.0	10.3	14.7	18.6	22.9	24.0	21.7	16.0	12.1	6.5		
	10%	DB	3.3	4.4	8.3	13.7	18.6	21.7	25.7	27.2	24.1	18.4	13.5	7.1		
		MCWB	1.2	2.1	5.7	9.5	14.1	18.2	22.5	23.6	20.8	15.1	10.7	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.5	10.2	14.5	17.7	21.4	25.2	26.0	24.7	19.9	15.5	10.9		
		MCDB	8.0	10.3	12.3	17.9	21.0	23.6	28.1	28.9	27.8	21.6	17.0	12.0		
	2%	WB	4.1	5.5	8.7	12.6	16.6	20.4	24.4	25.3	23.4	17.9	13.8	8.7		
		MCDB	5.5	7.5	10.9	16.0	19.7	22.5	27.0	27.9	26.2	19.6	15.6	10.2		
	5%	WB	2.7	3.8	7.4	11.3	15.7	19.6	23.7	24.7	22.4	16.7	12.4	6.7		
		MCDB	4.1	5.8	9.6	14.2	18.8	21.7	26.0	27.2	24.8	18.9	14.6	8.5		
	10%	WB	1.5	2.3	5.8	10.2	14.8	18.8	23.0	24.2	21.4	15.5	10.9	4.9		
		MCDB	3.1	4.3	8.1	12.9	17.8	20.9	25.0	26.6	23.6	17.8	13.1	6.7		
(35) Mean Daily Temperature Range	MDBR		3.9	4.4	5.7	7.2	7.2	6.5	5.7	6.3	6.9	6.7	5.7	4.6		
	5% DB	MCDBR	5.9	7.0	8.6	9.3	9.2	8.0	7.0	7.1	7.8	7.8	7.8	7.4		
		MCWBR	5.3	6.1	6.7	5.9	5.5	4.9	4.0	4.1	4.7	5.5	6.7	6.7		
	5% WB	MCDBR	5.9	6.9	8.2	7.9	8.0	6.6	5.8	6.0	7.0	7.0	7.5	7.3		
		MCWBR	5.6	6.5	7.0	5.9	5.5	4.4	3.5	3.8	4.6	5.7	7.2	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.383	0.425	0.493	0.538	0.553	0.525	0.534	0.493	0.435	0.419	0.412	0.397		
	taud		2.055	1.951	1.852	1.804	1.848	2.004	2.051	2.180	2.294	2.198	2.107	2.044		
	Ebn at Noon		728	754	746	745	746	769	758	778	802	761	693	671		
	Edh at Noon		116	150	185	208	203	174	165	141	118	116	111	110		
(44) All-Sky Solar Radiation	RadAvg		1.23	2.02	3.30	4.61	5.08	5.15	4.57	4.67	3.88	2.88	1.63	1.11		
	RadStd		0.10	0.22	0.26	0.47	0.53	0.47	0.46	0.45	0.31	0.19	0.16	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page